< HTML >
NOTES

LEARN HTML STEP BY STEP

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<HTML> <HEAD> <BODY>

< INTRODUCTION TO HTML >

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What is HTML?

<HTML> stands for HyperText, Markup Language

- ✓ Standard language for creating web pages.
- ✓ HTML documents are files ending with .html..

Full Form:

- ✓ <HTML>: HyperText Markup Language
- ✓ HyperText: Text linking to other text (hyperlinks)
- ✓ Markup: Symbols / tags to define elements
- ✓ Language: It's a coding language.

History of HTML

- ✓ Developed by Tim Berners-Lee in 1991.
- ✓ Launched as the first version of HTML (HTML.1.0)
- ✓ Evolved with HTML2.0, HTML4.0, XHTML, HTML5..

Uses of HTML

- ✓ Creating web pages & web apps.
- ✓ Embedding text, images, videos, and links.
- ✓ Building the structure of web pages.

HTML vs Web Page (in short)

- ✓ HTML is the code, a web page is how it looks in the browser.

<HTML> / <head> / <body> <h1> <p> <a> <img> <ul>

HTML vs Web Page (in short)

HTML is the code, a web page is how it looks in the browser.

<html> <head> <body> <h1> / <p> / <a> / <img> / <ul> <li>

< BASIC HTML STRUCTURE >

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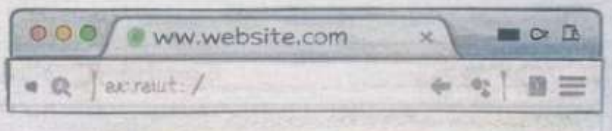
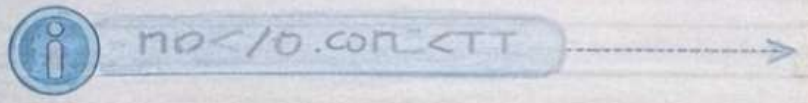
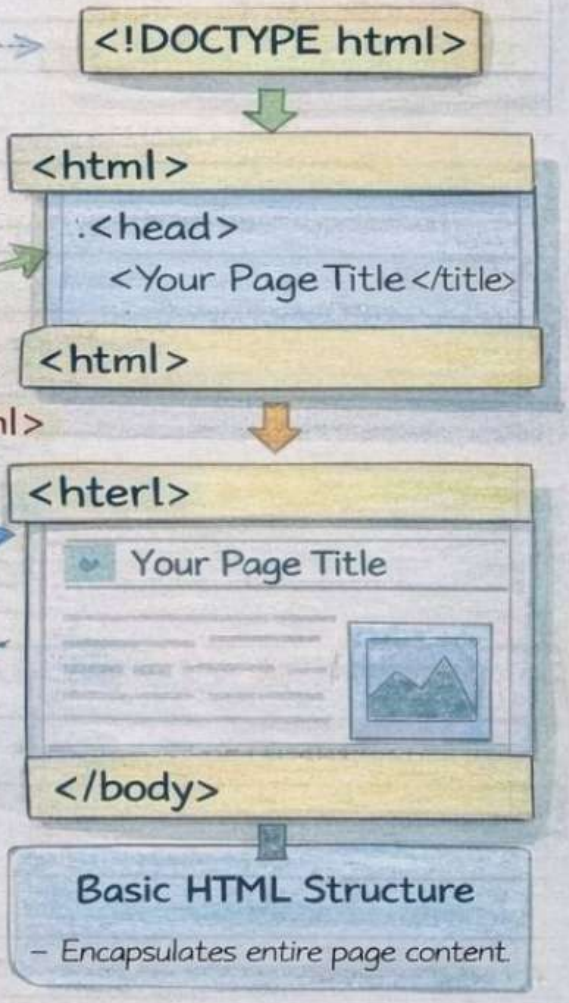
<!DOCTYPE html>

- Declares the document as HTML5.
- Always at the top.







- ✓ Declares the document as HTML5.
- ✓ - Always at the top.

<html> Tag

- ✓ Encloses the full HTML document.
- ✓ Starts with <html> and ends with &html>

<head> Tag

- ✓ Contains metadata and settings.
- ✓ Includes the <title> with page title displayed in browsers title bar.
- ✓ Other meta tags, CSS, etc.

<html> Tag

- ✓ Contains the visible content of the web page.
- ✓ Includes headings, paragraphs, images, links.
- ✓ Essentially, everything displayed to the user.

Core HTML Tags

<html> <head> <title> <body> <h1> <p> <img>

< HTML ELEMENTS & TAGS >

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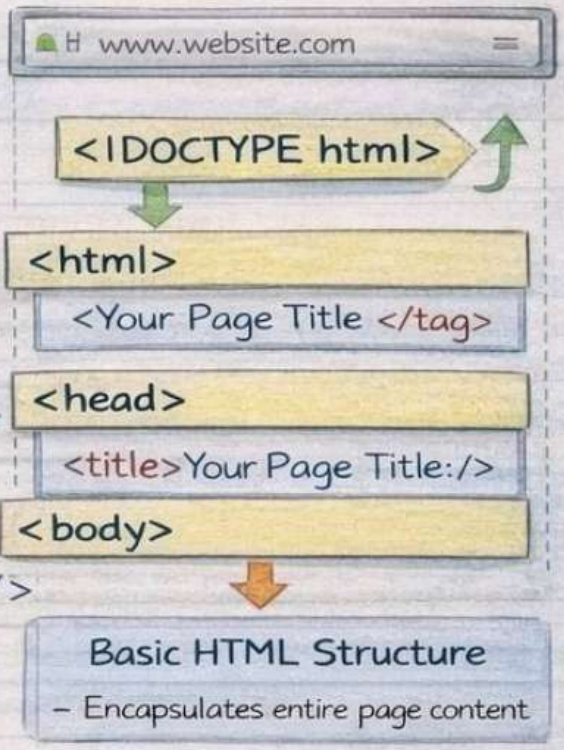
What are HTML Elements?

- ✓ Building blocks of web pages.
- ✓ Defined by HTML tags.
- ✓ Elements have opening & closing tags.



HTML Tags

- ✓ Provide additiond into'.<(lctml>.
- ✓ Ex ample: <title> Sign =Icongs"sg>
- ✓ <a hiref="https//example.com">

  <img src="image.jpg" alt="Description" />

Common HTML Tags

- ✓ <html> The root tag of an HTML document.
- ✓ <head> Contains metadata and links to CSS.
- ✓ <body> Contains the main content of the web page.
- ✓ Defines a paragraph.
- ✓ <a> Defines a hyperlink.
- ✓ <img> UL: Creaté ar"uunordered (bulleted) list.
- ✓ <ol> = ol> Creates anordered (numbered) list.

Common HTML Tags

- ✓ <html> The root tag of an HTML document
- ✓ <head> Contains metadata and links to CSS. <hi>: Paragraph
- ✓ <body> Contains the main content of the web page.
- ✓ Defines a paragraph.

<a href="https://example.com">Visit Example/a>

<img src="logo.png" alt="Website Logo'/>

 <a img <ul> <li>
<hi>

< TEXT FORMATTING TAGS >

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<h1> <h6> Tags

✓ Define headings from largest to smallest:

```
<h1.Main heading</hi1>
<h6>Small heading</h6>
```

<h1.Main heading
<h1,Main heading
<h1,Small heading
<h6,Small heading
<h6.,Small heading

<p> Tag

✓ Defines a paragraph of text:

```
<p> This is a paragraph of text. :: <p>
```

This is a paragraph

Small heading

<b> Tag

✓ Makes text bold:

```
<b> Bold text</b>
```

Bold text

Italic text.

<r> Tag

✓ Makes text italic:

```
<u>Underlined text <iu></lu>
```

Underlined text.

 Tag

✓ Inserts a line break:

```
Line one.<br> Line two...
```

Line one.
 Line two...

Line one.

<hr> Tag

```
<a href="https//example.com"> Visit. Exemple </in>
<img src="Tago.png" <lt="Website Logo: </>
```

<p><e><img><ul><li><ht>

< LISTS IN HTML >

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Ordered List

- ✓ Creates a numbered list.
- ✓ Uses the tags <ol> and <li>.

```
<ol>
 <li>First item</li>
 <li>Second item</li>
 <li>Third item</hb>
</ol>
```

→ Items are preceded by numbers.

1. First item
2. Second item
3. Third item
4. Small heading

<ol type="1'> | <ol type="A'> | <ol type="A'> | <ol type="I"><i">

- ✓ Optional "type" attribute changes list numbering style.

Unordered List

- ✓ Creates a bulleted list.
- ✓ Uses the tags <ul>and ll>.

```
<ul> Item One </li>
<ul> Item Two </li>
<ul> Item Three </li>
```

- ✓ Items are preceded by bullets.

1. First item.
2. Second item.
3. Third item.

This is a paragraph of text.

1. First item.

Line one. ---- Line two..

Description List

- ✓ Creates a list of terms with descriptions.
- ✓ Uses the tags <l>,<ul> and <ld>.

```
<dt>HTML</dt>
        Language for creating web pages
</dt> Item One</t>
</dt> Item Three</l>
```

HTML
· Language for creating web pages.
· CSS.
· Styles the HTML content

Definition term + description.

<ol> <ul> <di> <li> <dt> <dd> <hr>

< LINKS & IMAGES >

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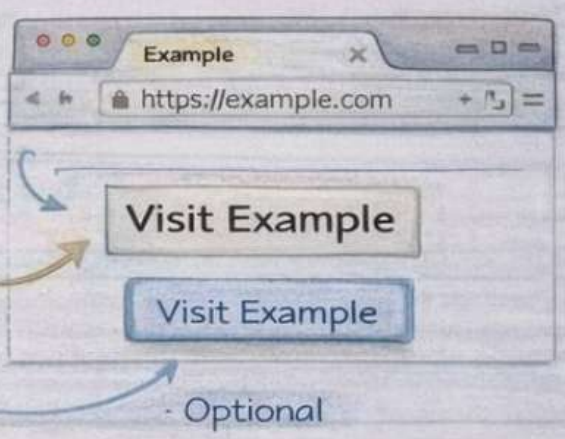
Links <a> Tag)

✓ Defines hyperlinks (clickable links).

✓ Uses the <a> tag with href attribute




```
<ol>
  <a href="https://example.com"
    target="_blank">
</ol>
```

✓ href: Specifies URL of the link.

✓ target="_blank">: Open the link in a new tab.

Images <img> Tag)

✓ Embeds an image on the page.

✓ Uses the <img> tag (self-clo-

```
<img src="logo.png" alt="Website Logo"> width="600" height="200" />
```

✓ Items are preceded by bullets.

Images <img> Tag)

✓ Embeds an image on the page.

✓ Uses the <img> tag (self-closing).

✓ Uses the src and alt attributes.

```
<img src="logo.png" alt="Website Logo" width="600" height="200" />
```

✓ Width and height: Sets the dimensions of the image.

✓ - Always use meaningful alt text.

< TABLES >

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What is a Table?

- ✓ Organizes data into rows & columns.
- ✓ Uses the <aable>tag with nested tags:

```
<a href="https://example.com"
 target="_blank>
```

Basic HTML Table

Name	Age	Country
John	25	USA
Sarah	28	UK

Basic Table Tags

- ✓ <table> Creates an HTML table
- ✓ <tr> Defines a table row.
- ✓ <th> Defines a table header (bold + centered)
- ✓ <td> Defines a table data cell.

Sample HTML Table Structure

- ✓ <table border="1">.../table>
  - Adds border to the table.
- ✓ <td <olspan="2">...</td>
- ✓ <td rowspan="2">....</td>
- ✓ <td width ad height:
  - Sets the tables width and height.

tabole

https://example.com

```
<tr>
  <table>
    <tr>
      <th Name</th>
      <th Country</th>
    </tr>
   <tr>
     <td>John</td>
     <td>25</td>
    </tr>
     <td>Sarah</td>
     <td>28 </td>
   </tr>
</table>
```

Borders & Attributes

- ✓ <table border="1">... < table>

  <td colspan="2">...</td> - Spans across 2 columns.

  <td rowspan="2">...</td> - Spans across 2 rows.

- ✓ <table width="500" height="200">.. table>-Sets the table's width

Core Table Tags: <ta><th><td>

< HTML FORMS >

What is a Form?

- ✓ Collects user input on a webpage.
- ✓ Uses the <form> tag to create a data entry area.
  - Inputs inside the form tag.

```
<a  form
    <form action="/submit.php" method="post">
    + – Form Elements (inputs) go here —
</form>
```

Login Form

Username:

JohnDoe

Password:

•••••••

Submit

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- ✓ action: Specifies where the form data is sent,
- ✓ method: Defines the HTTP method (get or post).

Common Input Types & Tags

- ✓ <input type="text"> Text field.
- ✓ <input type="password"> Password field.
- ✓ <input type="email"> Valiates email address.
- ✓ <input type="checkbox> Checked ← ☑
- ✓ <input type="radio"> Select one option.
- ✓ <input type="submit"> Sumit submit button.

table

https://example.com

Name	Age	Country
John	25	USA
Sarah	28	UK

Key Input Attributes

- ✓ name: Identifies the input (sent to server).
- ✓ value: Sets the initial value of the input.
- ✓ placeholder-: Provides a hint for the user.
- ✓ required: Makes the input mandatory.

<form> <input> <select> <textarea> <button>

< HTML FORMS INPUT TYPES >

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What is a Form?

✓ Collects user input on a webpage.

✓ Uses the <form> tag to create a data entry

- Inputs inside the form tag.

Form Type

Username: JohnDoe

Password: •••••

Submit

```
<form
  <input type="text" placeholder="Enter text here">
</form>
```

✓ action: Specifies where the form data is sent.

✓ method: Defines the HTTP method (get or post).

Common Input Types & Tags

✓ <input type="text"> Text field.

Name: Enter text here

Table

https://example.com

Quantity: 1 to 10 1 to 10

✓ <input type="checkbox" checked>

☑ Accept Terms. 1 to 10

✓ <input type="email'> name="gender" value="male">

Gender: ○ Male ◉ Female

✓ <input type="radio">

Message: Enter your message here

✓ <input type="submit"> Submit button.

Select (Dropdown)

✓<select><option>Option 1</option>

Country:

Submit Button

✓ <input type="submit">

Click Me!

Submit Button

✓ <table onclick="alert'Clickedl"> </button>

Button (General)

Click Me!

<input> <select> < textarea> < button>

< HTML MEDIA TAGS >

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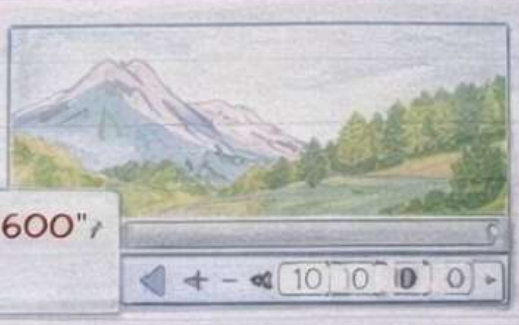
Video <video> Tag)

- ✔ Embeds a video on the page.
- ✔ Use the <video> tag with src attribute:

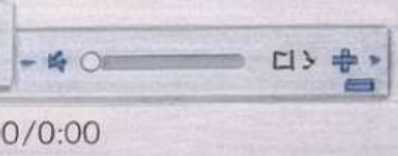


```
<video src="video.mp4" controls width="600"/
<<video>
```

Audio <audio> Tag)

- ✔ Embeds audio (sound) on the page.
- ✔ Use the <audio> tag with src attribute.

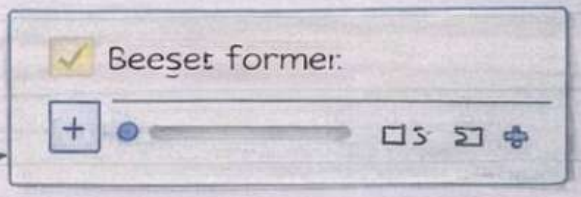
```
<audio src="audio.mp3" controls><Audio>
```



0:00/0:00

Images <<img> Tag)

- ✔ Displays an image on the page.
- ✔ Use the <img> tag with src and + alt attributes.




```
<img src="photo.jpg" alt="Description of image" />
```

Select <Dropdown)

- ✔ Sblecs <<pwp> tag with src "and alt attributes.
- ✔ - Always use a meaningful alt attribute.

```
<img src="photo.jpg" alt="Description of image" />
```

< HTML TIPS & BEST PRACTICES >

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Video <Video> Tag)

- ✔ Start with a basic HTML structure.
- ✔ Use the <Video> tag with src attribute:

```
<!DOCTYPE html>
<html>
  <title> Page Title </title>
</head>
```

Comments

- ✔ Use comments <!-...- > to document your code.

Indentation

- ✔ Properly indent your code for readability.

```
<div class="container">
  <ul>
    <li> List item </li>
  </ul>
```

Semantic Elements

- ✔ Use semantic elements to give meaning to your content:

```
<header> <nav> <main> <article> <aside><section><footer> ...
```

Accessibility

- ✔ Always use a meaningful alt attribute for images.

```
<img src="logo.png" alt="Site Logo" />
```

- W3C Validator: (validator.w3.org)
- HTMLHint: (htmlhint.com)

Validation Tools

- ✔ Use online validators to check your HTML code.
  - W3C Validator: (validator.w3.org)
  - HTMLHint: (htmlhint.com)

Core Semantic Elements:

<header><nav><main><article><aside><section><footer>

< HTML REVISION >

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Basic Structure

- ✓ Start with a basic HTML structure.
- ✓ Use the <Video> tag with src attribute.

```
<!DOCTYPE html>
<html>
  <title> Page Title</title>
</head>
```

Comments

- ✓ <H1 Mäin Heading</h1>
- ✓ <P This is a paragraph <p>

Text Formatting

- ✓ Headings: <h1> to <h6>
- ✓ Paragraphs: <p>
- ✓ Bold <b>,) & Italics <i,

Links & Images

- ✓ Hyperlinks: <a href="<httpseanle.Com>> Visit Example</a>
- ✓ Open link in new tab: target="_blank"
- ✓ Images: <img src="path/to/image.jpg" alt="Description"

Tables & Forms

- ✓ Tables: <table><tr,<th,<td>
- ✓ Forms: <form> <input> <select><textarea>
- ✓ Input types: text, password, email, number, etc.
- ✓ Input attributes: name, required, placeholder, etc.

CSS Selectors:

<element> .class, #id> <section> <section><footer>